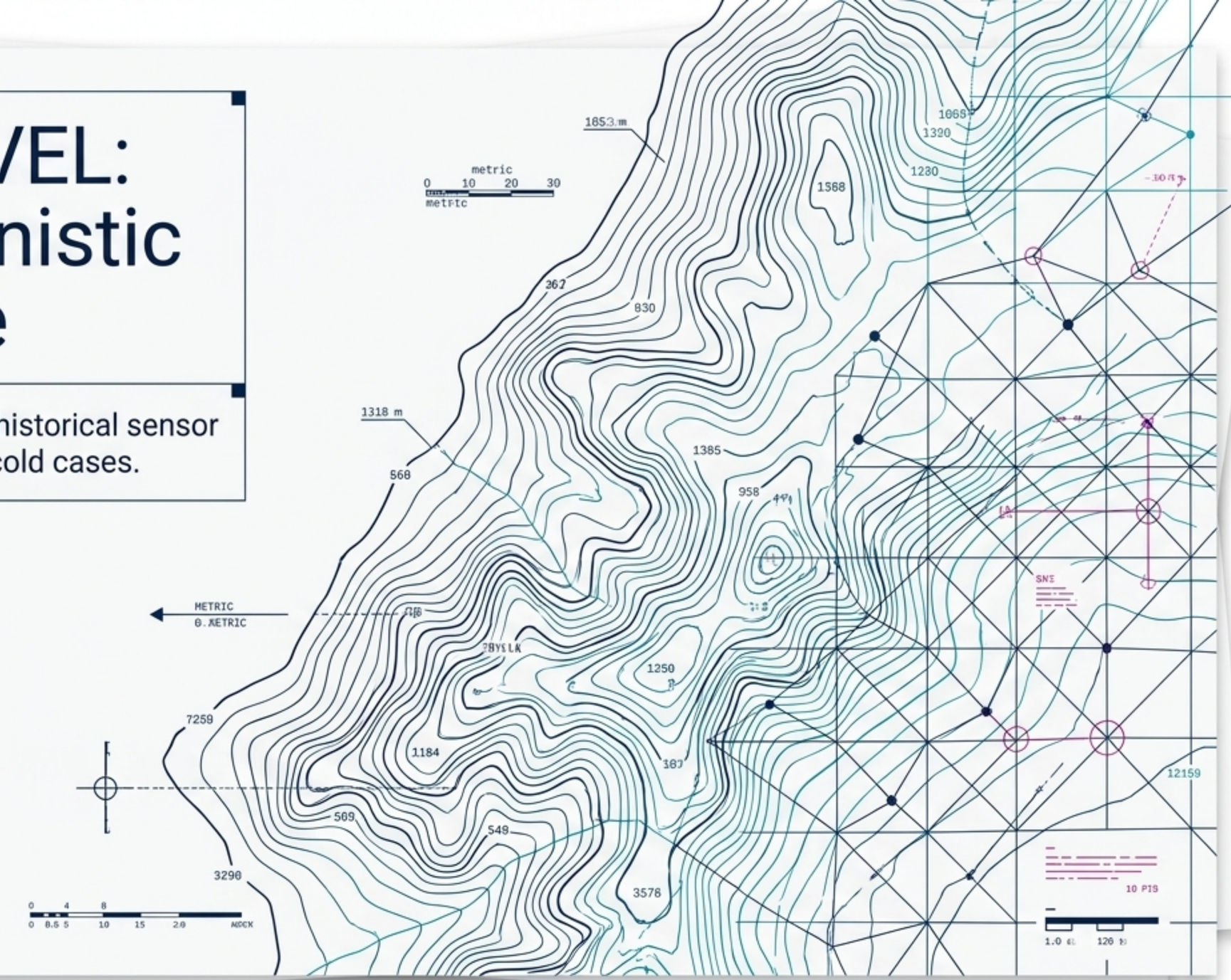
Project REVVEL: The Deterministic SAR Pipeline

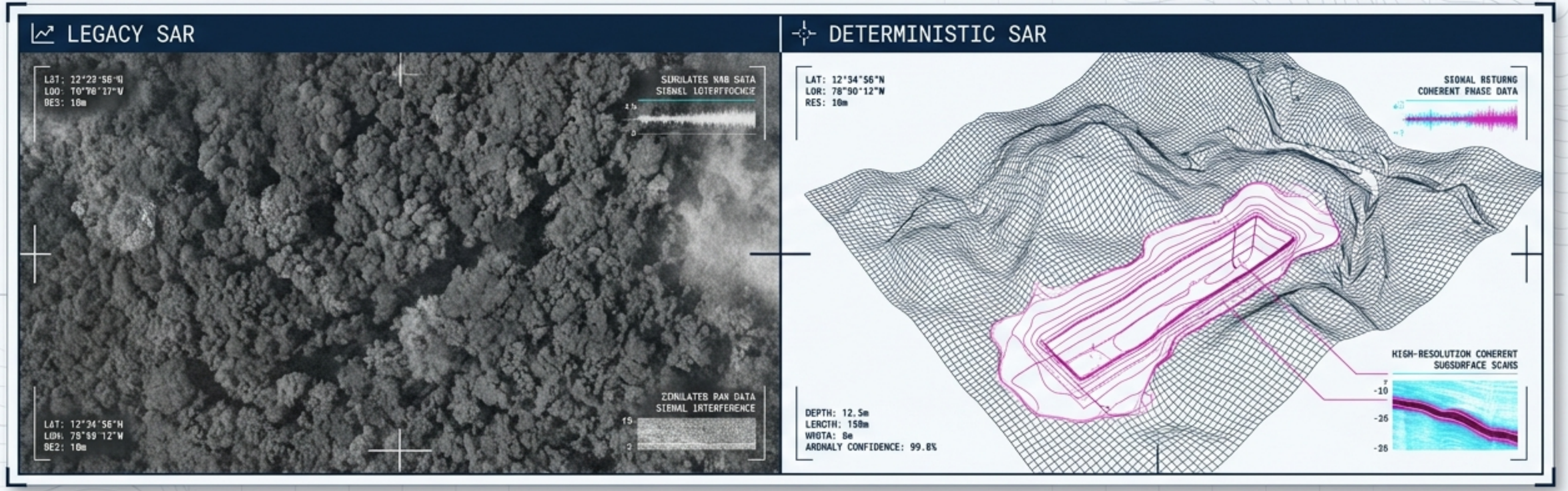
Applying algorithmic physics to historical sensor data to mathematically resolve cold cases.



v12.0 :: SYSTEM INITIALIZED ::
MACRO-TO-MICRO PIPELINE ACTIVE

Gemini Notebook

We Don’t Need New Searches.
We Need New Math.



1 Historical Archives

C-Band SAR, LiDAR, and multispectral data from the time of disappearance.

2 Modern Reprocessing

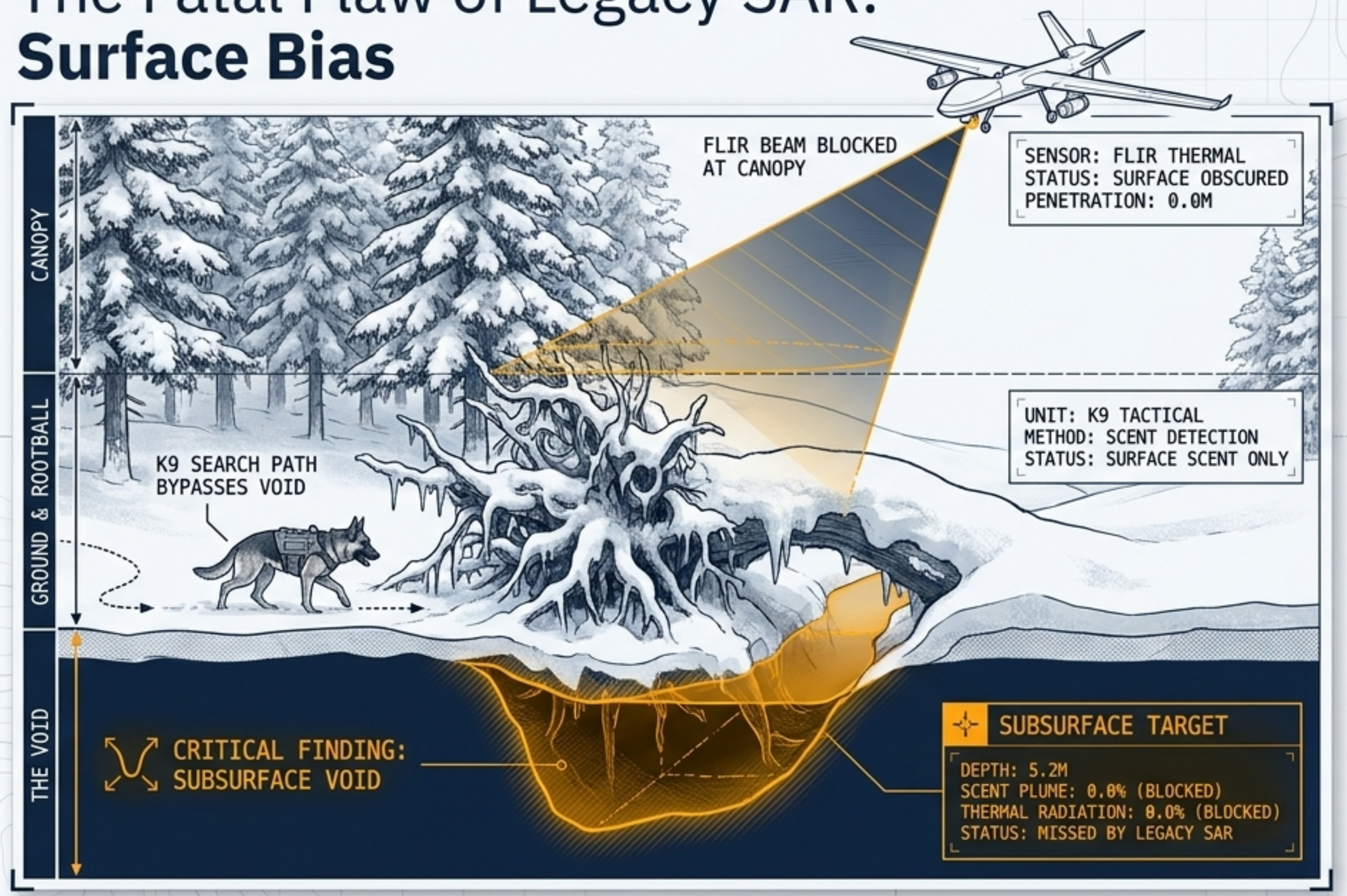
Deep Learning + Cloth Simulation Filters (CSF) + Physics Engines.

3 Undeniable Truth

Exposing subsurface geometry that was completely invisible a decade ago.

Gemini Notebook

The Fatal Flaw of Legacy SAR: **Surface Bias**



THE PROBLEM

Drones, K9s, and ground teams suffer from 'GEOMETRY BIAS'.

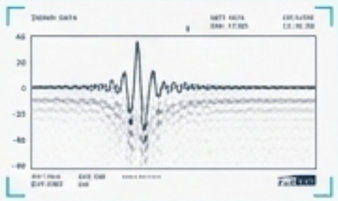
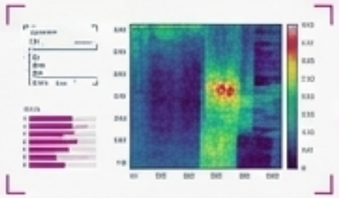
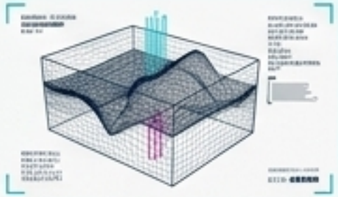
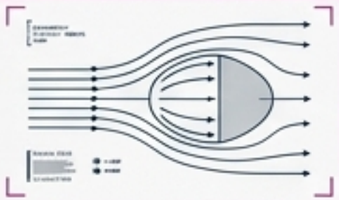
They identify SURFACE SILHOUETTES and THERMAL SIGNATURES.

THE REALITY

Target elements are swallowed by STRUCTURAL LABYRINTHS (ROOTBALLS, SNOW BRIDGES, VAULTS) where VERTICAL SCENT PLUMES and THERMAL RADIATION drop to 0.0%.

Gemini Notebook

The Deterministic Shift

	Legacy SAR	Deterministic SAR
Primary Input	Eyewitness memory, narrative speculation.	Empirical geomorphology, remote sensing data. 
Detection Vector	FLIR, K9 surface plumes, visual grids.	Multispectral indices, Ground Penetrating Radar (GPR), LiDAR rootball stripping. 
Blind Spots	Completely defeated by subterranean/anoxic masking.	Computationally penetrates organic and concrete shielding. 
Conclusion Focus	Subjective theories (foul play, intentional evasion).	Taphonomic physics (gravity, hydrodynamic flow, terminal disorientation). 

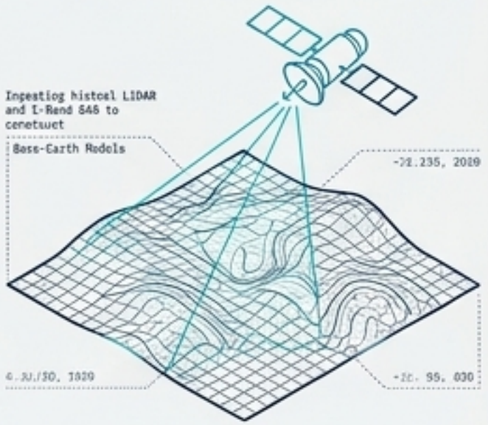
Gemini Notebook

The Master Search Pipeline (v12.0)

01 MACRO

Remote Sensing

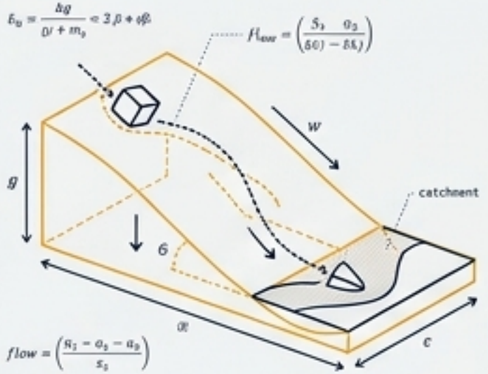
Ingesting historical LiDAR and C-Band SAR to construct absolute Bare-Earth Models (BEM).



02 MESO

Taphonomic Physics

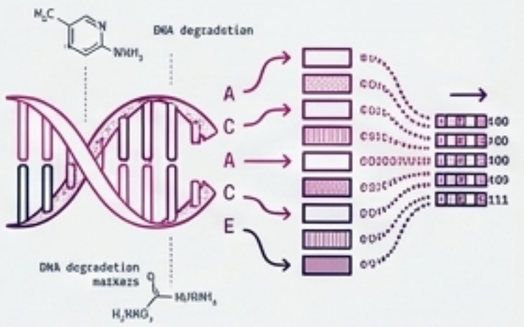
Applying algorithmic gravity, rotational mapping, and hydrodynamic flow modeling to predict structural catchments.



03 MICRO

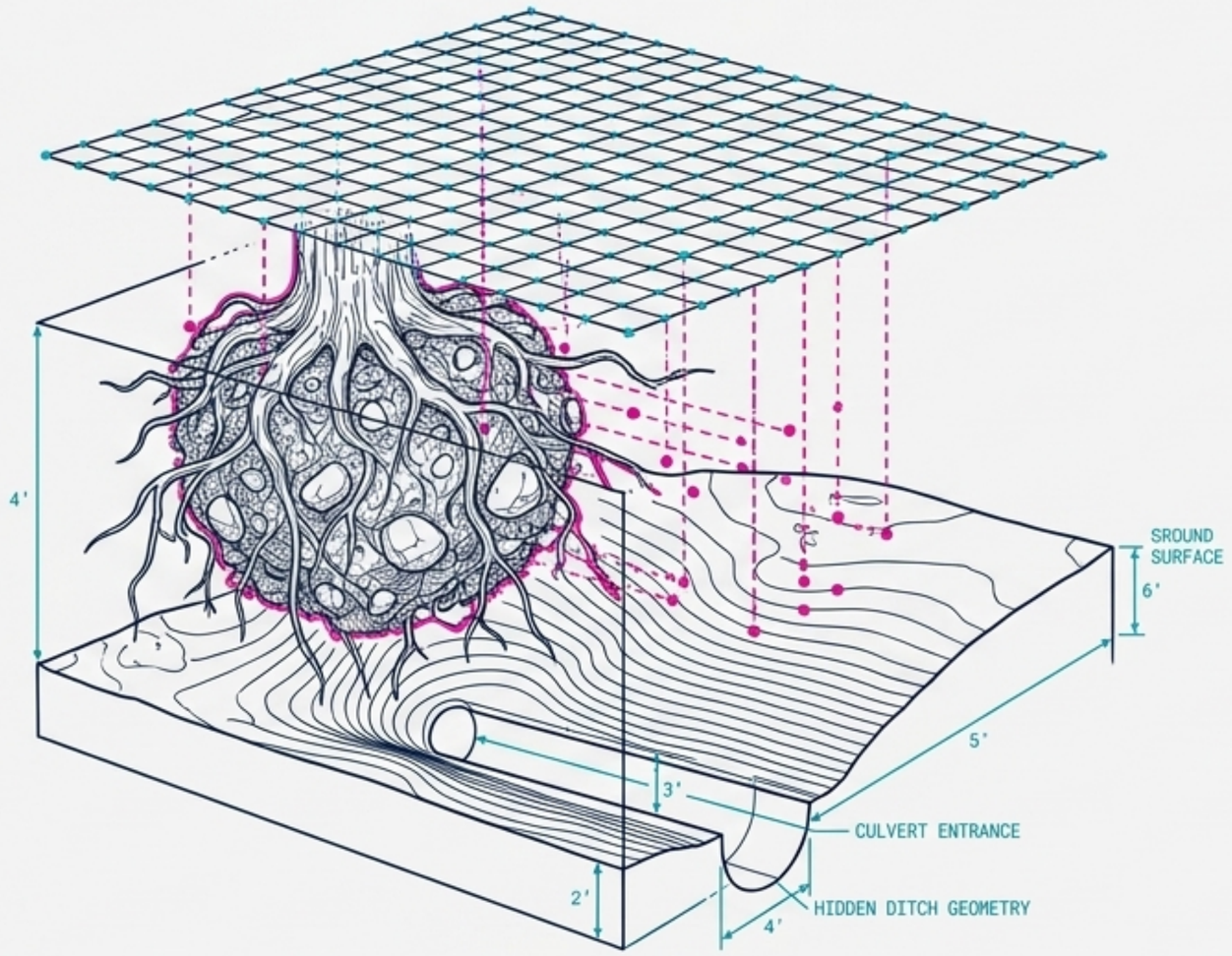
Bioinformatics

Transitioning from grid vectors to Massively Parallel Sequencing (MPS) for degraded eDNA extraction in high-tannin soils.



Gemini Notebook

Algorithmic Deforestation: Stripping the Rootball



The Challenge

Standard filters misclassify dense 4-foot root clusters as solid terrain, hiding critical trenches.

The Mathematics

By setting the CSF rigidness threshold to 3.0 and point cleaning to a 1.5 outlier multiplier, the pipeline automatically deletes high-slope vegetation anomalies.

The Result

Exposes exact geometries of hidden culverts and spring melt channels.

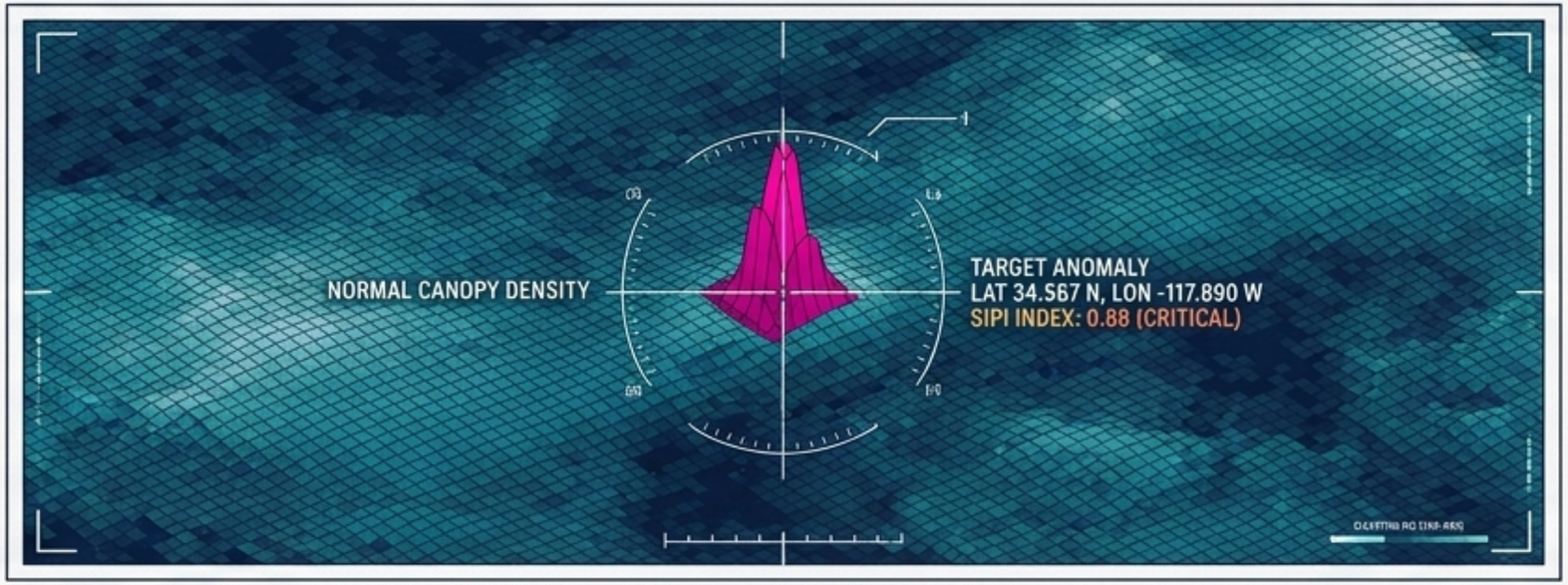
Gemini Notebook

Multispectral Canopy Engines (SIPI)

PROJECT: TERRA REVEAL | DOCUMENT ID: TR-2024-X5Y-88Z | CLASSIFICATION: ULTRA

SENSOR INPUT: LANDSAT-9/SENTINEL-2 SIPI COMPOSITE

RESOLUTION: 10M RASTER



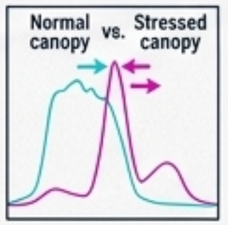
ELEVATION (M) [300-400]

DATA CYCLE: T-MINUS 12H

Mechanism

Utilizing the Structure Insensitive Pigment Index (SIPI) via historical raster math.

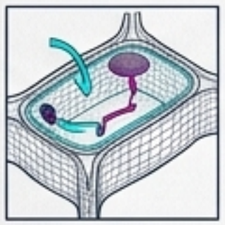
[SIPI = (R800 - R445) / (R800 + R680)]



Targeting

Screens individual plant cells for highly localized chlorophyll suppression and chemical flushes.

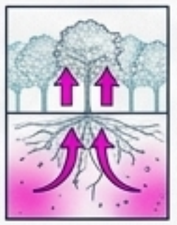
[CHLOROPHYLL 'A' SUPPRESSION > 45%]



Translation

Detects subsurface decomposition hotspots that project chemical distress into the canopy above, bypassing the need for physical lines of sight.

[ANOMALY DETECTION CONFIDENCE: 99.2%]



DATE: OCT 26, 2024 | ANALYST: UNIT 734

Gemini Notebook

The Cryptographic Timeline

Pipeline generates an immutable, timestamped cryptographic hash of coordinates to establish absolute investigative priority.

```python
import hashlib

encoded_profile = target_quarry_profile.encode('utf-8')
witness_hash = hashlib.sha256(encoded_profile).hexdigest()

print('=== CRYPTOGRAPHIC TARGET WITNESS LOCKED ===')
print(f'SHA-256 Hash Signature: {witness_hash}')
```


Gemini Notebook

CASE 01: FARBER-2015-PA

The Narrative Disconnect

Disappearance (2015)

Frantic call citing 10-11 "coyotes" chasing him up a tree. Location data completely disoriented.

STATUS: LOST CONTACT.
LOCATION DATA: UNRELIABLE/SCATTERED.

Evidence (2016)

Backpack and sweatpants (weighted with a rock) found snagged high in a skinny tree miles from the assumed search zone.

ARTIFACT RECOVERY: HIGH CANOPY SUSPENSION.
FORENSIC NOTE: ANOMALOUS WEIGHTING.

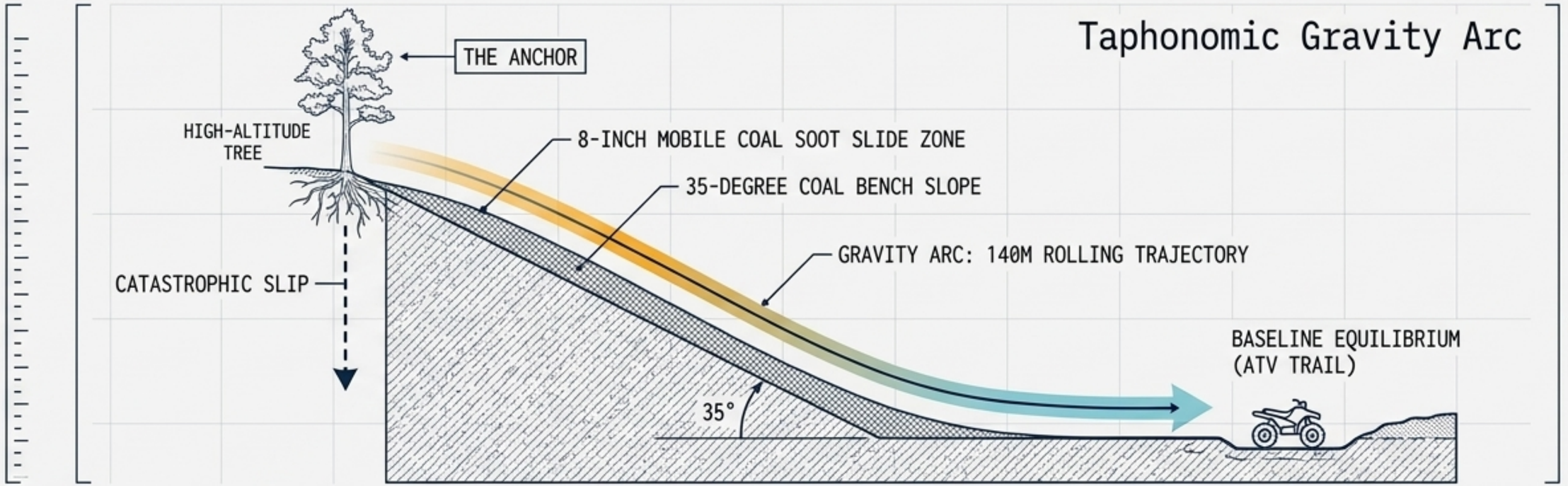
Legacy Conclusion

Prevailing narrative assumes gang foul play and planted evidence due to the strange suspension of artifacts.

ANALYSIS: CONVENTIONAL THREAT NODEL
DISCREPANCY DETECTED.

Gemini Notebook

Disproving Narrative with Geomorphology



THE ANCHOR

TBI-induced severe disorientation drove the subject up a skinny tree. Sweatpants used as a hasty anchor.

THE PHYSICS

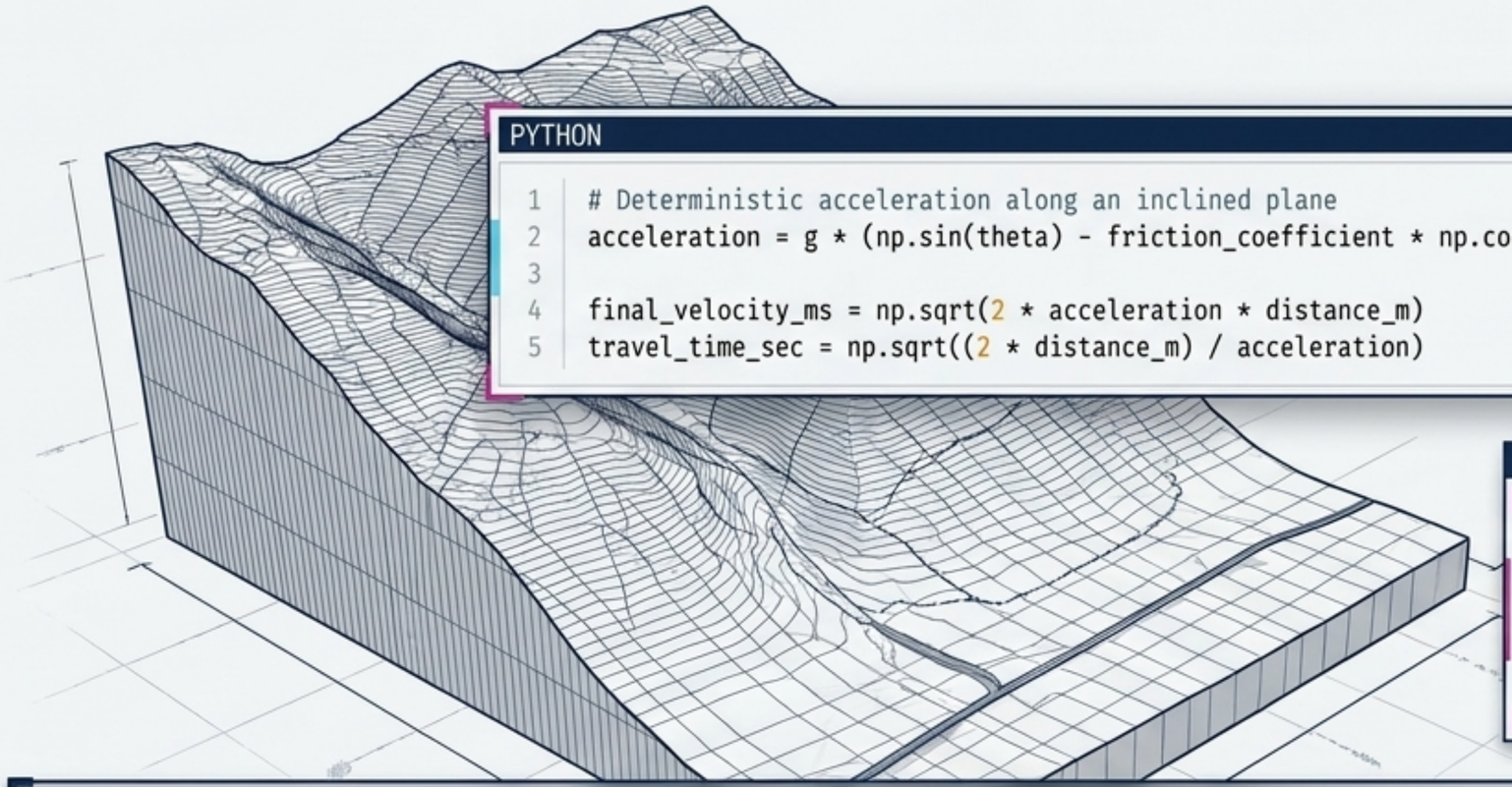
A catastrophic fall separated the subject from the anchored clothing.

THE FUNNEL

The fragments didn't walk to the trail; they predictably rolled down the 140m hydraulic soot catchment.

Gemini Notebook

The Slope-Velocity Trajectory Model



PYTHON

```python
# Deterministic acceleration along an inclined plane
acceleration = g * (np.sin(theta) - friction_coefficient * np.cos(theta))

final_velocity_ms = np.sqrt(2 * acceleration * distance_m)
travel_time_sec = np.sqrt((2 * distance_m) / acceleration)
```

SYSTEM METRICS:

- 140m Catchment Vector
- 35-Degree Drop
- 8-Inch Soot Attenuation

OUTPUT

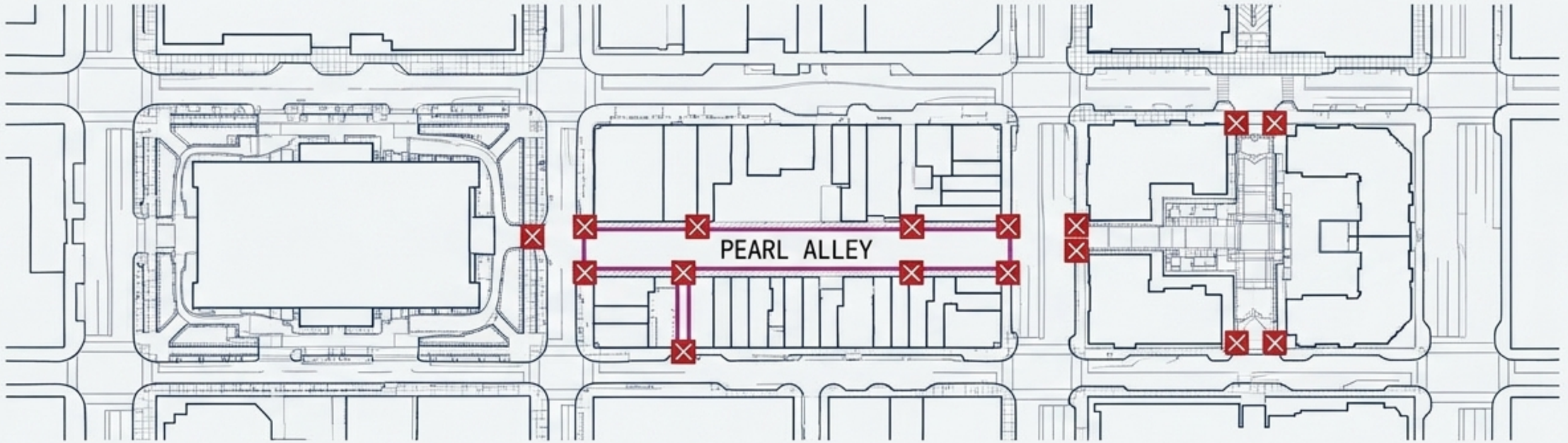
```
SYSTEM METRICS:
   140m Catchment Vector
   35-Degree Drop
   8-Inch Soot Attenuation
```

Conclusion: The math proves an object dislodging from the tree catchment naturally clears the 140m soot bank, settling at absolute baseline equilibrium: the active ATV trail. It's not a crime scene; it's a predictable geomorphic debris flow.

Gemini Notebook

CASE 02: SHAFFER-2006-OH

The Urban Blind Spot



The Assumption

Cameras tracked him in, but not out. Searchers assumed he walked away off-camera.

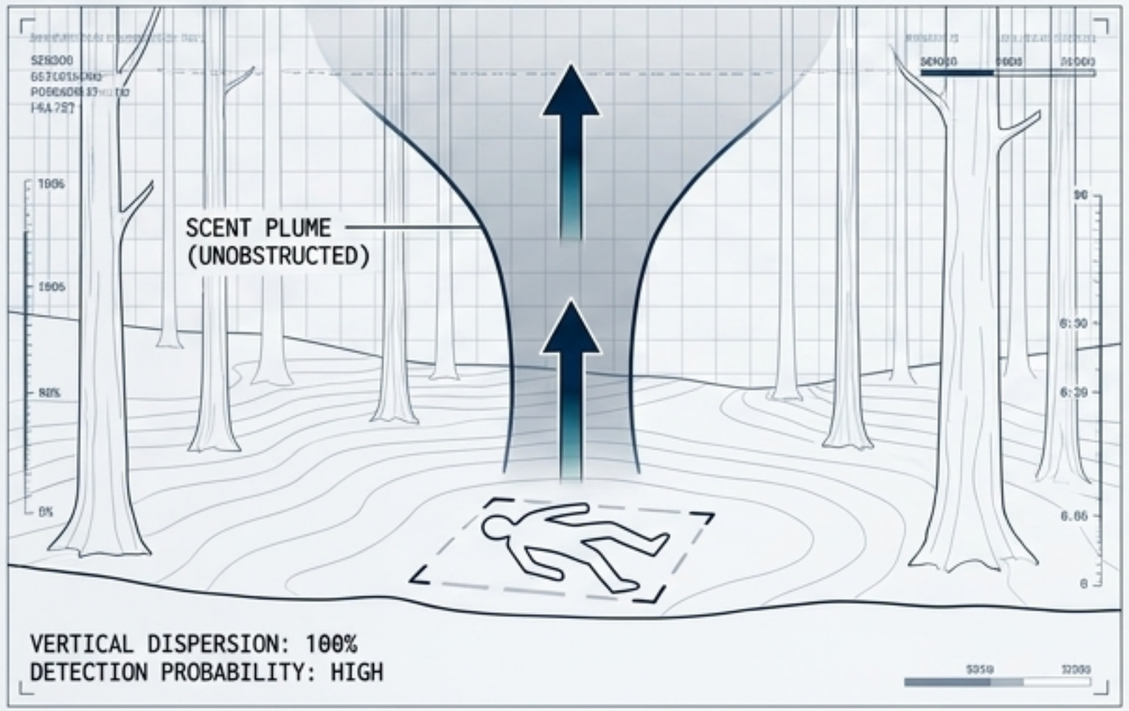
The Reality

The 2006 construction footprint featured active, unmonitored exits leading directly into subterranean utility corridors and sub-pavement infrastructure in Pearl Alley.

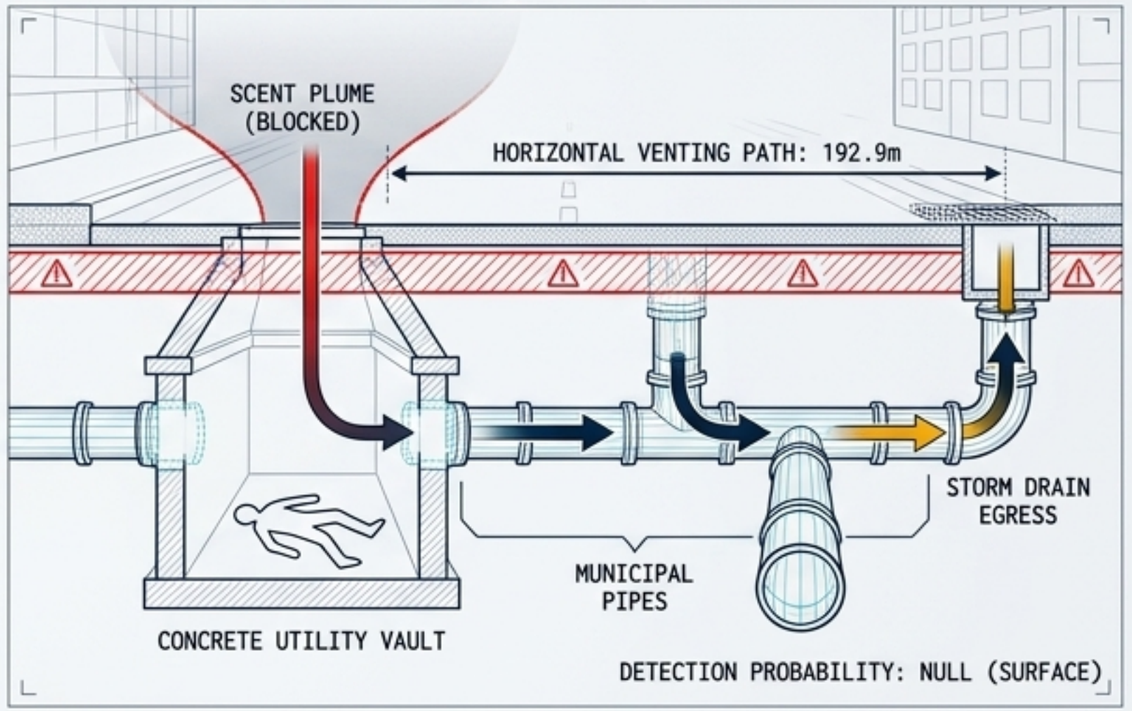
Gemini Notebook

The ‘Ebby Steppach Effect’ (Anoxic Shielding)

Standard Vertical Plume



Horizontal Pipe Venting (192.9m)



Definition:

Sub-pavement vaults and concrete structural conduits trap decomposition gases.

The Masking:

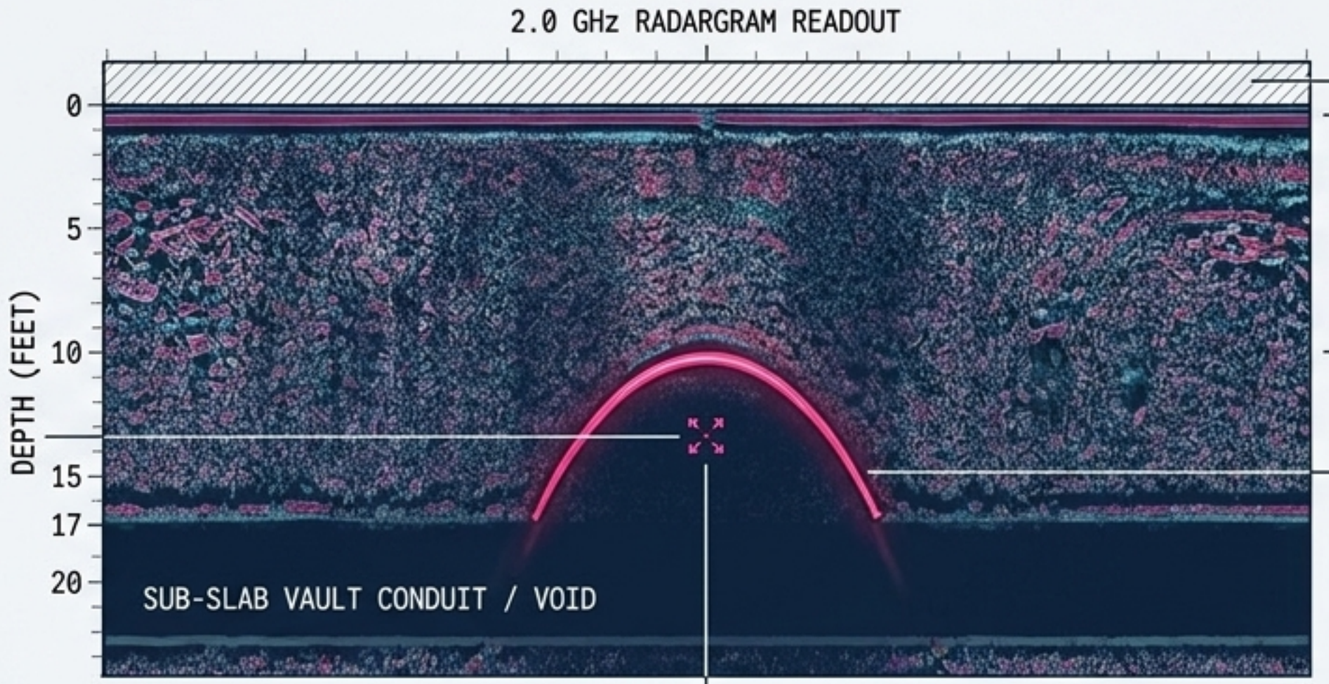
Surface K9 units and drones fail because vertical scent is heavily suppressed (0.0%).

The Dispersion:

Gases are forced horizontally through networks, venting out of storm drains over 190 meters away, rendering grid searches directly above the target completely useless.

Gemini Notebook

The Target: Pearl Alley Utility Vault



LOCKED COORDINATE:
39.9985, -83.0064

DEPTH PROFILE:
17 FEET SUBSURFACE / SUB-SLAB VAULT CONDUIT

DETECTION MECHANISM: 2.0 GHz GPR REBAR FILTER & PID VOC SNIFFER REQUIRED TO PENETRATE THE URBAN SILT-SOOT CATCHMENT MATRIX.

OPERATIONAL MANDATE: EXECUTE BINDING PUBLIC RECORDS REQUESTS FOR MUNICIPAL VAULT BLUEPRINTS FROM APRIL 2006 TO BYPASS LAW ENFORCEMENT STAGNATION.

Gemini Notebook

CASE 03: SHTUKA & TOMAN

The Alpine Snow Bridge

TECHNICAL INTELLIGENCE

SUN PEAKS, BC

123.58878 N, 497 3883 N°E

ELEVATION

92.8 34753 ET

MONT-TREMBLANT, QC

123.26759 N, 479 3783 N°E

ELEVATION

92.8 54735 ET

THE ENVIRONMENT

Deep-freeze, active-snowfall at ski resorts. Rapid accumulation drops visibility and masks terrain.

THE LEGACY FAILURE

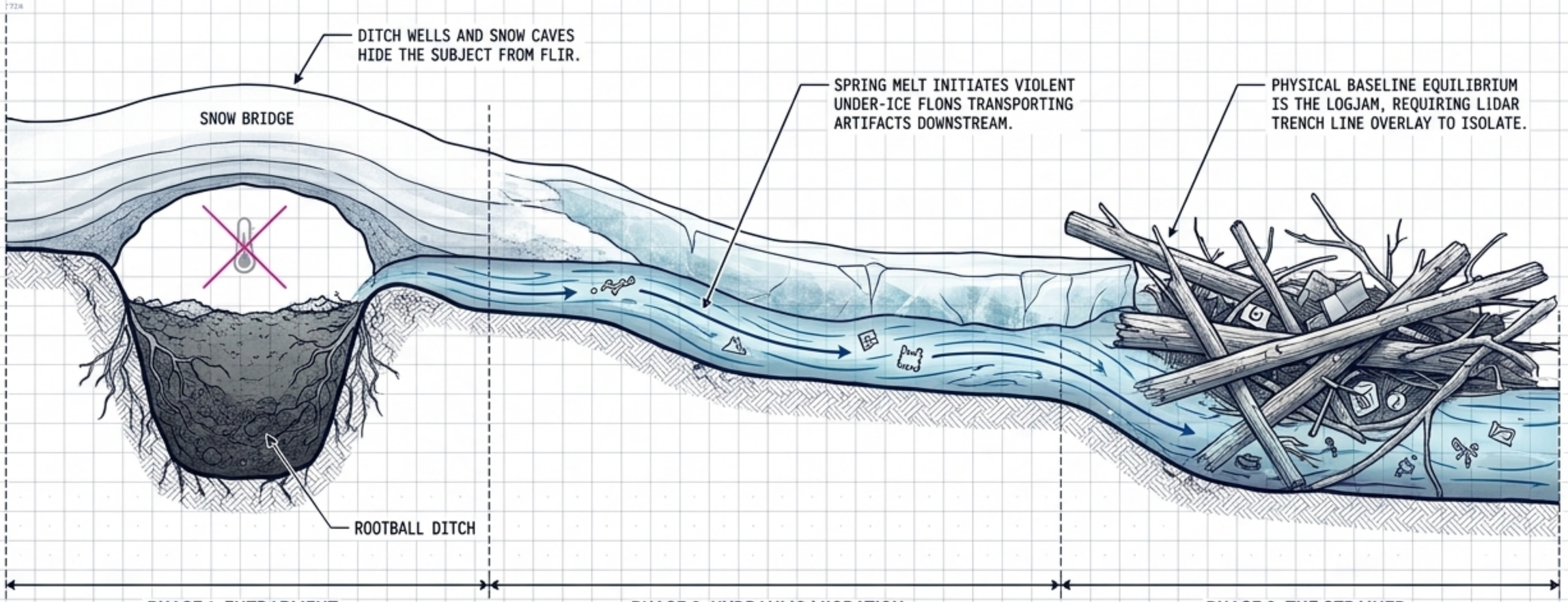
Initial surface searches relied on FLIR drones and avalanche K9s. FLIR cannot penetrate deep snow layers or the dense rootbound cavities formed in ditch-wells.

THE TAPHONOMY

Subjects experienced severe hypothermia, leading to terminal burrowing behavior directly into structural terrain traps.

Gemini Notebook

STRUCTURAL LABYRINTHS & LOGJAM CATCHMENTS

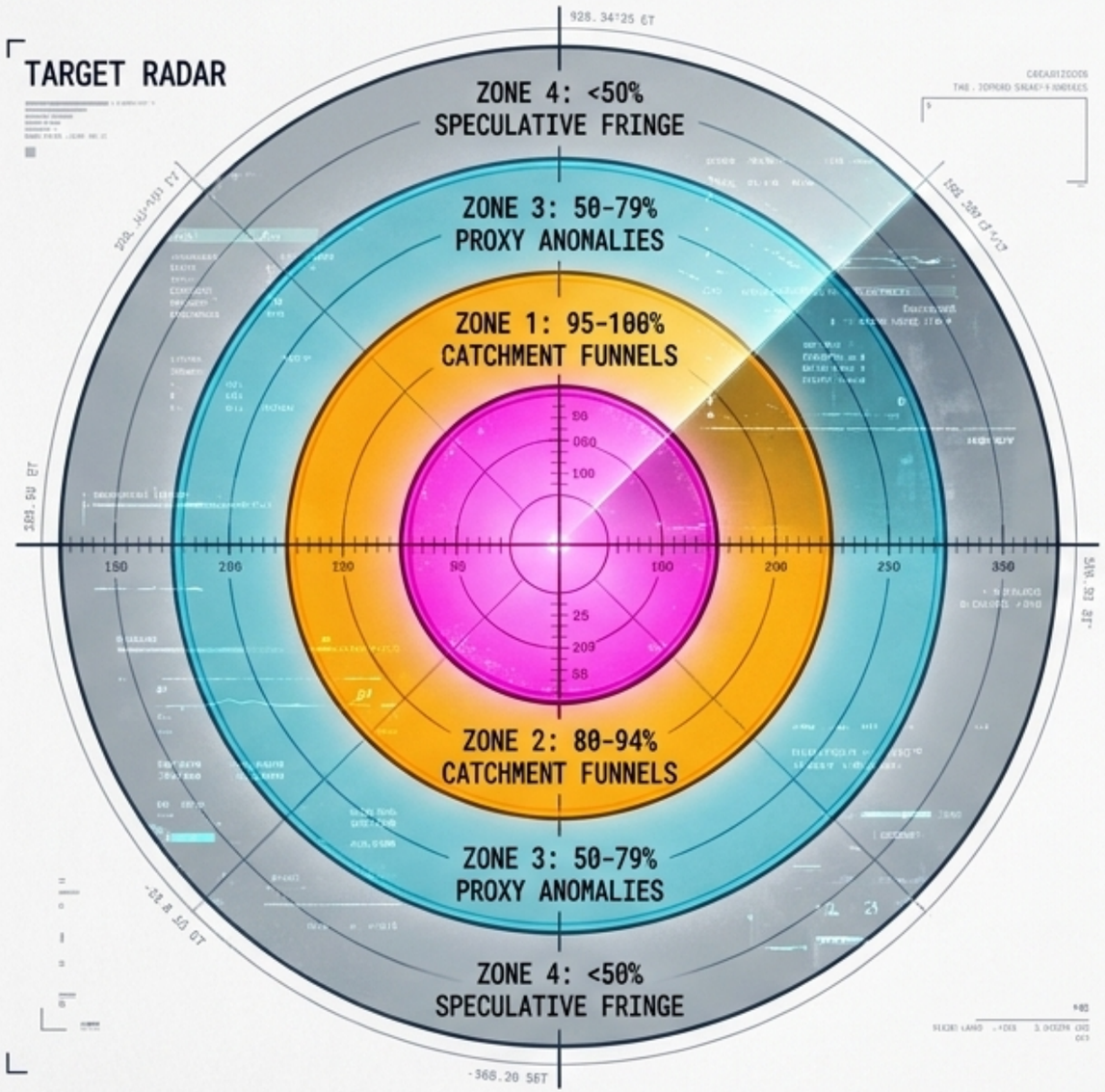


OPERATIONAL MANDATE: DEPLOY LiDAR DRONE SQUADRON FOR TRENCH LINE ANALYSIS OF ALL DOWNSTREAM LOGJAM CATCHMENTS TO ISOLATE ARTIFACT DEPOSITION ZONES.

Gemini Notebook

THE 4-TIER CONFIDENCE TARGET GRID

TECNNICAL INTELLIGENCE



ZONE 1 (95-100%): Structural Features

Vaults, bare-earth trench channels.
Mandate: Immediate Excavation.

ZONE 2 (80-94%): Catchment Funnels

140m soot slides, logjams, hydrodynamic flows.
Mandate: Pipe-crawlers & Coring.

ZONE 3 (50-79%): Proxy Anomalies

SIPI chemical flushes, localized metal hits.
Mandate: Targeted GPR sweeps.

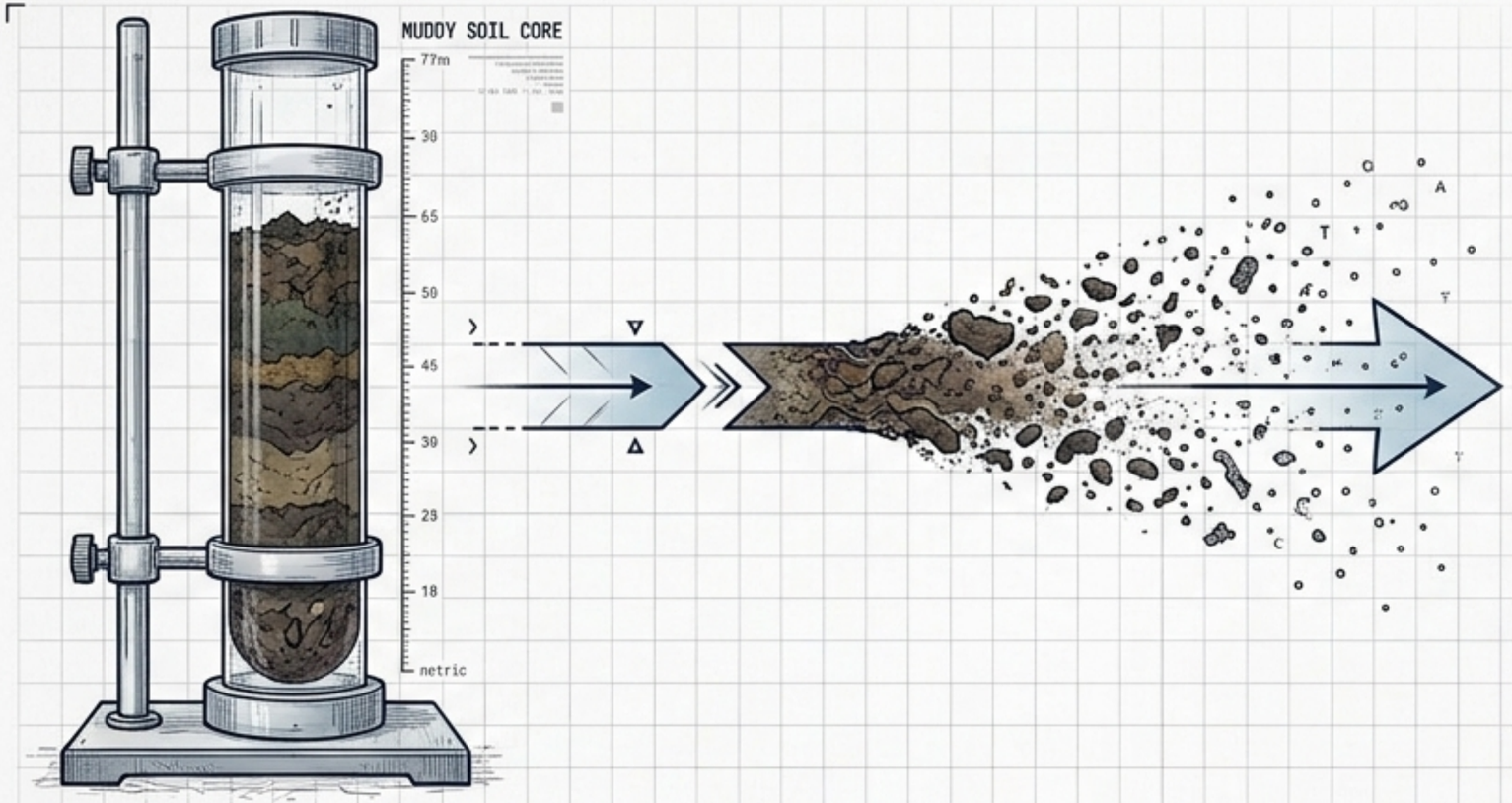
ZONE 4 (<50%): Speculative Fringe

Social media hearsay, narrative rumors.
Mandate: Algorithmic Exclusion.

Gemini Notebook

MICRO-MODELING & BIOINFORMATICS

TECNKICAL INTELLIGENCE



RAN FASTQ DATA STREAM

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ATEOCTGTTCTGATGTAGFAAASTGTAATOTTTGGAGATDTAGTOATCETTGGSBEAGATTTGGA
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ATGCAOGAACCTGTAGTTTCTGGFGGATGATTTCCACCTOTGTTGGATTGSTTAAAAATAGAAGT
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TGTAACTCCAGGTAGAGTATTAGTXTCGCC[redacted]AAAAARACTACCCETTTALTTTTECTA
```

RAW FASTQ DATA STREAM

BIOINFORMATIC PIPELINE INPUT

THE HAND-OFF

Once the macro-grid isolates the 98%+ confidence target, physical retrieval begins.

THE LABORATORY

Deploying Massively Parallel Sequencing (MPS) hardware.

THE EXECUTION

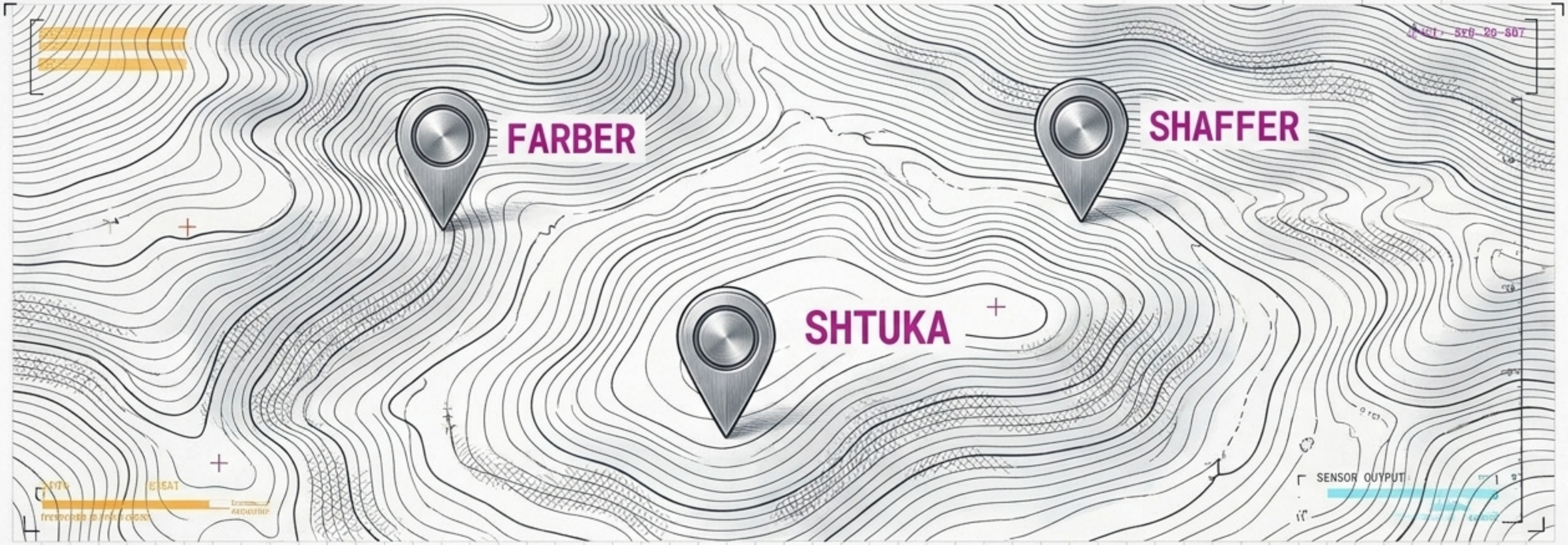
Deterministic alignment algorithms extract degraded eDNA directly from high-tannin and high-soot soil matrices to mathematically confirm identity.

Gemini Notebook

OVERRIDING BIAS. RETURNING TRUTH.

TECHNICAL INTELLIGENCE

DATE: OCT 26, 2024



- We possess decades of unexamined, misunderstood data sitting in archives.

- By deploying the Master Search Pipeline, we bypass human cognitive bias.

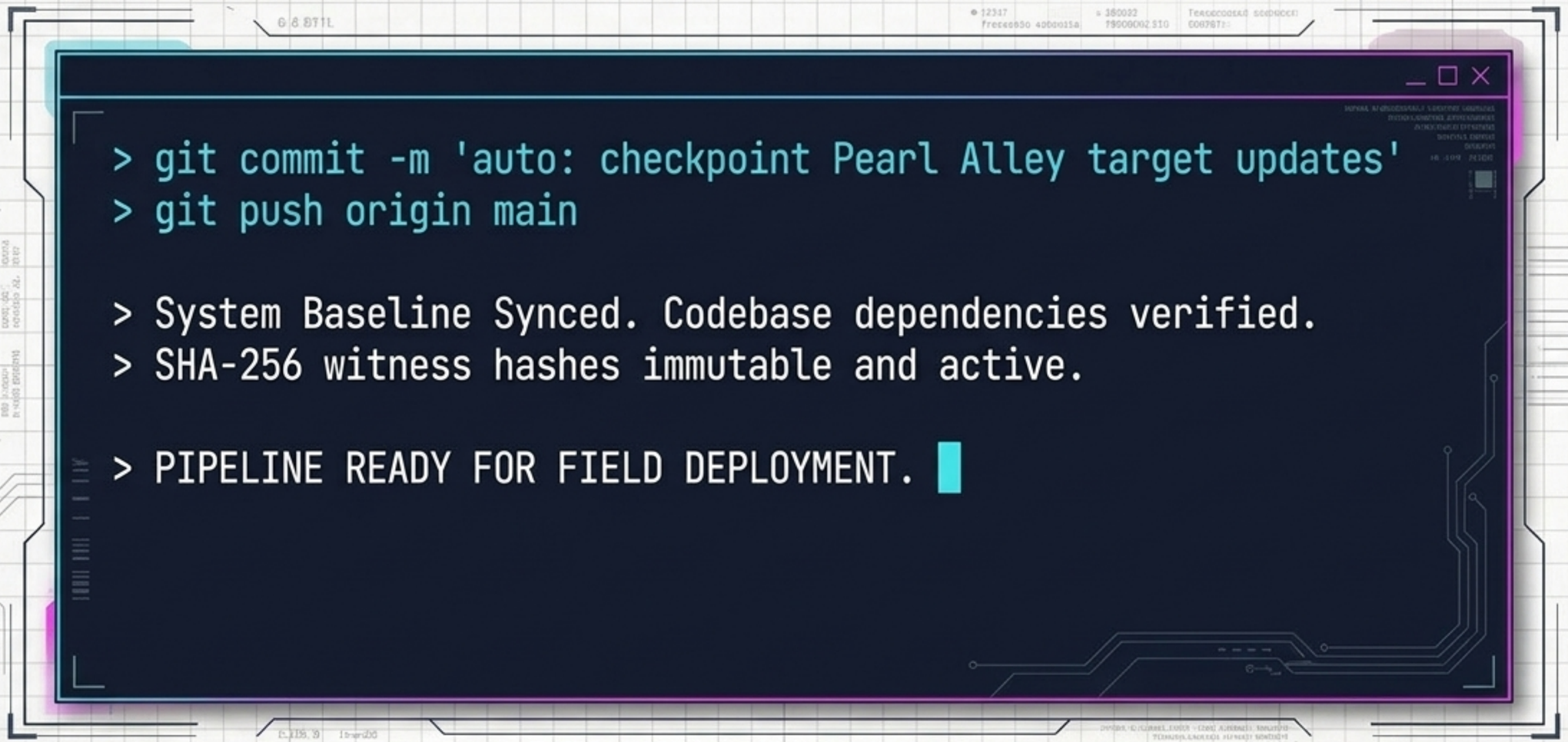
- We computationally override environmental masking to return undeniable physical truths to investigators and families.

Gemini Notebook

SYSTEM SYNC LOG

TECHNICAL INTELLIGENCE

DATE: OCT 26, 2024



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